

Program: Diploma in Integrated Circuits Design & Fabrication	
Course Code: 6371B	Course Title: Sensors and Actuators
Semester: 6	Credits: 4
Course Category: Program Elective	
Periods per week: 4 (L:4 T:0 P:0)	Periods per semester: 60

Course Objectives:

- To introduce the basic principles of different types of sensors, transducers and actuators.
- To learn different transducers and their basic applications.
- To know the basic principles of micro actuators.

Course Prerequisites:

Topic	Course code	Course name	Semester
Basic principle of resistance, inductance, capacitance, diode, transistors etc.	2031	Fundamentals of Electrical & and electronics	2
Permeability, Electromagnetic induction, self and mutual inductance; Transformer.	2031	Fundamentals of Electrical & and electronics.	2

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Explain the basics of transducers & sensors.	15	Applying
CO2	Explain different types of inductive, capacitive and piezo electric transducers.	14	Understanding
CO3	Explain photo electric transducers, radiation detectors, ultrasonic detectors, Hall effect transducers.	14	Understanding

CO4	Explain the concepts of actuators, electrical actuators and micro actuators.	15	Understanding
	Series Test	2	

CO – PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	2						
CO2	2						
CO3	2						
CO4	2						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Explain the basics of transducers & sensors		
M1.01	Explain transducers, sensors and classify them.	6	Understanding
M1.02	Classify different types of resistive transducers	6	Understanding
M1.03	Explain the applications of different types of resistive transducers.	3	Understanding
Contents: Introduction to Transducers, Resistive transducers: Define transducer and Sensor; classification of transducers as active-passive, analog-digital, unidirectional - bidirectional, mechanical – electrical, primary - secondary transducers. Resistive Transducers: Basic principle of resistive transducers; Working of linear potentiometer, loading effect; Basic principle of strain gauge, different types, derivation of equation for gauge factor, Construction and working of Light dependent resistor (LDR), Resistance temperature detector (RTD) and Thermistor; Displacement measurement using potentiometer.			
CO2	Explain different types of inductive, capacitive and piezo electric transducers.		
M2.01	Explain the basic principle of inductive transducers and classification of inductive transducers.	3	Understanding

M2.02	Explain the construction and working of LVDT and eddy current transducer and LVDT for pressure measurement.	4	Understanding
M2.03	Explain basic principle of capacitive transducers and working of different types of capacitive transducers and capacitive level gauge.	4	Understanding
M2.04	Explain working principle of piezo electric transducers and piezo electric pressure transducer.	3	Understanding
	Series Test – I	1	

Contents:

Inductive, Capacitive and Piezo electric transducers:

Basic principle of inductive transducer; Classification of inductive transducers; Construction and working of Linear Variable Differential Transformer (LVDT); pressure measurement using LVDT; basic principle of Eddy current transducers.

Illustrate basic principle of capacitive transducer; Working of different types of capacitive transducers (variable area, variable distance, variable dielectric type); Level measurement using capacitive transducer.

Basic principle of piezo electric transducer; Equation for voltage sensitivity and charge sensitivity of the piezo electric transducer; Equivalent circuit of piezo electric transducer; Pressure measurement using piezo electric transducer.

CO3	Explain photo electric transducers, radiation detectors, ultrasonic detectors, Hall effect transducers.		
M3.01	Explain photoelectric transducers- photo conductive (photo diode, photo transistor), photo emissive transducers (photo multiplier tube), solar cell and photo voltaic cell.	5	Understanding
M3.02	Explain the basic principle of Ultrasonic transducer and level measurement application.	4	Understanding
M3.03	Explain the construction and working principle of Hall effect transducer and current measurement application.	5	Understanding

Contents:

Photoelectric, Ultrasonic and Hall Effect Transducers: Illustrate photoelectric transducers-Working principle of photo conductive -photo diode, photo transistor- transducers. Working principle of photo emissive -photo multiplier tube- transducers, Working principle of solar cell and photo voltaic cell.

Working Principle of Ultra Sonic transducer. Construction and working principle of Hall effect transducer. Application of Ultrasonic transducer for level measurement, Hall Effect Transducers for current measurement.

CO4	Explain the concepts of actuators, electrical actuators and micro actuators		
M4.01	Explain the concept of actuators.	3	Understanding
M4.02	Explain different electrical actuating systems.	6	Understanding
M4.03	Explain basic principles of micro actuators & its different types.	6	Understanding
	Series Test – II	1	

Contents:

Actuators: Definition, types (basics only)-electric linear, electric rotary, piezoelectric, selection criteria of actuator, Applications.

Electrical Actuating Systems: Solid-state switches, Solenoids, DC motors, Stepper motors, Servo motors.

Micro Actuators: Actuation principle, shape memory effects-one way, two way and pseudo elasticity. Types of micro actuators- Electrostatic, Magnetic, Fluidic, Inverse piezo effect, other principles.

Text / Reference:

T/R	Book Title / Author
R1	A.K. Sawhney, “A course in Electrical and Electronic measurement and Instrumentation”, Dhanpathrai and Co.
R2	D.V.S. Murthy, “ Transducers and Instrumentation”, PHI. Pvt. Ltd.
R3	D.Patranabis, “ Principles of Industrial Instrumentation”, Tata McGraw-Hill. Education,
R4	J.B.Gupta, “Electrical and Electronic Measurements and instrumentation”, Kataria Pulishers.
R5	Ranganathan S, “ Transducer engineering ”, Allied publishers.
R6	Krishna Kant, “Computer-Based Industrial Control”, PHI.

R7	D. Patranabis, “Sensors and Transducers”, PHI.
R8	Massood Tabib and Azar, “Microactuators Electrical, Magnetic, thermal, optical, mechanical, chemical and smart structures”, First edition, Kluwer academic publishers, Springer, 1997.
R9	Manfred Kohl, “Shape Memory Actuators”, first edition, Springer

Online Resources:

Sl.No	Website Link
1	https://www.youtube.com/watch?v=FYggY7LLDrw
2	https://www.slideshare.net/manash234/classification-of-transducers
3	https://www.electronics-tutorials.ws/io/io_1.html
4	https://www.egr.msu.edu/classes/ece445/mason/Files/4-Sensors_ch2.pdf
5	https://www.youtube.com/watch?v=XSbUnUADlno
6	http://www.kyowa-ei.com/eng/technical/strain_gages/principles.html
7	https://www.youtube.com/watch?v=anCnrtjNLQM
8	http://www.te.com/content/dam/te-com/documents/sensors/global/lvdt-tutorial.pdf
9	https://www.electronics-tutorials.ws/electromagnetism/hall-effect.html
10	https://www.brighthubengineering.com/hvac/53147-how-capacitive-transducers-work/
11	https://www.powershow.com/view4/556e94-ODlIO/TRANSDUCERS_VARIABLE_RESISTIVE_CAPACITIVE_INDUCTIVE_powerpoint_ppt_presentation
12	https://www.ctscorp.com/resource-center/tutorials/piezo-basics/
13	https://www.youtube.com/watch?v=CckMERCuXvE
14	https://www.youtube.com/watch?v=7jQ12uUqK6E